Sensitivity to Neural Network Tuning

BZAN 554 Group Assignment 2 · Group 6

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Study objective and approach

- Examine how tuning parameters affect prediction accuracy.
- Change one parameter at a time, comparing each setting with the same baseline.
- Reset to the baseline for each experiment; improvements are not carried forward.

Data and preprocessing

- One row represents a product's selling episode on the website.
- Predict **quantity** from SKU, price, order, duration, and category.
- Random row split: **70% training / 15% validation / 15% test**, seed 42.
- Fit min-max scaling and category/SKU mappings on training data only.
- Scale numeric inputs and quantity, one-hot encode category, and embed SKU in eight dimensions.
- Load the CSV in chunks of **16,384 rows** for incremental training.

Repeated SKUs can occur across splits. The comparison concerns held-out episodes, including episodes for products seen in training.

Training and evaluation

- Every logged run trains for **20 epochs**, using random seed **42**.
- Shuffle training observations within each chunk and update by mini-batch.
- Evaluate validation MSE after each epoch and restore the best weights.
- Keep the learning rate constant within a run. No early stopping.
- Report validation metrics on **75,053 observations**. RMSE and MAE use the supplied quantity scale.

Source: We tested a total of 31 configurations ran on both October 30 and 31. Scores match across dates. Tables use October 31 throughout, including runtime.

These repeated runs use the same seed and do not measure variability across seeds. The CSV contains validation results only.

Baseline model

Three sigmoid hidden layers: **21 / 12 / 7 neurons**. Eight-dimensional SKU embedding, one-hot category, scaled numeric inputs, and a linear output.

Setting	Value
Optimizer / learning rate	Adam / 0.001
Batch size	1,024
Selected epoch	19

Validation R^2	MSE	RMSE	MAE
0.3852	1,612.48	40.16	21.40

All following ΔR^2 values compare with this baseline. Positive values indicate improvement.

Batch size

Batch size	Validation R^2	RMSE	ΔR^2	Seconds
32	0.5210	35.44	+0.1358	187.4
128	0.5277	35.20	+0.1425	61.2
256	0.4304	38.65	+0.0452	39.6
512	0.4102	39.33	+0.0250	28.5
1024 (baseline)	0.3852	40.16	+0.0000	22.9

Finding: Batch 128 has the highest validation R^2 in this comparison, but takes about 2.7 times the baseline runtime.

Interpretation: Smaller batches make more weight updates per epoch. This comparison includes that extra optimization work. Runtime covers preprocessing and the training loop, including validation.

Number of hidden layers

Hidden layers and neurons	Validation R^2	RMSE	ΔR^2
1: 21	0.3667	40.76	-0.0185
2: 21 / 12	0.3591	41.00	-0.0261
3: 21 / 12 / 7 (baseline)	0.3852	40.16	+0.0000
4: 21 / 12 / 7 / 4	0.4023	39.59	+0.0171
5: 21 / 12 / 7 / 4 / 2	0.2513	44.31	-0.1339

Finding: Four layers improve R^2 slightly. Five layers perform substantially worse.

Interpretation: Extra depth does not guarantee better predictions. The narrow added layers also change capacity, so this result applies to these specific architectures.

Neurons per hidden layer

Three-layer architecture	Validation R^2	RMSE	ΔR^2
21 / 12 / 7 (baseline)	0.3852	40.16	+0.0000
8 / 8 / 8	0.3884	40.05	+0.0032
16 / 16 / 16	0.3909	39.97	+0.0057
32 / 32 / 32	0.3549	41.13	-0.0302
64 / 64 / 64	0.4145	39.19	+0.0293
128 / 128 / 128	0.2915	43.11	-0.0937

Finding: The 64 / 64 / 64 network performs best among the tested widths. The widest network performs worse than the baseline.

Interpretation: More neurons change capacity and optimization. Scores do not improve steadily with width. These trials also replace the baseline's taper with equal-width layers.

Hidden activation function

Activation	Validation R^2	RMSE	ΔR^2
Sigmoid (baseline)	0.3852	40.16	+0.0000
ReLU	0.4526	37.89	+0.0675
Tanh	0.3582	41.03	-0.0270
ELU	0.4057	39.48	+0.0205

Finding: ReLU gives the largest gain in this comparison. Tanh reduces R^2.

Interpretation: ReLU avoids sigmoid's positive-side saturation, which can help gradient flow. This is a possible explanation for the gain, not a measured gradient result.

The output layer remains linear for quantity regression.

Optimizer

Optimizer	Validation R^2	RMSE	ΔR^2
Adam (baseline)	0.3852	40.16	+0.0000
SGD	-0.1866	55.79	-0.5718
AdaGrad	-0.1838	55.72	-0.5690
RMSprop	0.1002	48.58	-0.2850

Finding: Adam performs best with the shared learning rate of **0.001**. SGD and AdaGrad have negative validation R^2.

Negative R^2 means greater squared error than predicting the validation-set mean.

Interpretation: Optimizers behave differently at the same learning rate. These runs compare a shared setting rather than each optimizer's best possible tuning.

Learning rate

Adam learning rate	Validation R^2	RMSE	ΔR^2
0.00001	-0.1105	53.97	-0.4957
0.0001	0.2903	43.14	-0.0949
0.0005	0.3969	39.77	+0.0118
0.001 (baseline)	0.3852	40.16	+0.0000
0.005	0.3626	40.89	-0.0226
0.01	0.4860	36.71	+0.1009
0.1	0.4783	36.99	+0.0931

Finding: 0.01 performs best among these rates. The smallest rate produces negative R^2.

Interpretation: A very small rate may learn too slowly within the shared training budget. Larger rates help here, but the relationship is not monotonic.

Sensitivity findings

Comparison	Highest observed setting	Validation R^2
Batch size	128	0.5277
Hidden layers	4: 21 / 12 / 7 / 4	0.4023
Hidden neurons	64 / 64 / 64	0.4145
Activation	ReLU	0.4526
Optimizer	Adam (baseline)	0.3852
Learning rate	0.01	0.4860

Learning rate and optimizer produce the largest observed spreads in R^2 across their tested settings.

Each row summarizes a separate comparison against baseline R^2 **0.3852**. These settings were not combined or confirmed as a final model.

Scope and limitations

- One seed and one split limit conclusions about repeatability.
- Equal epochs give smaller batches more updates. Runtime also varies.
- Validation selects the best epoch and supports these comparisons. A final test score is absent from the CSV.
- Independent comparisons do not measure interactions, such as ReLU with a different learning rate.
- Learning-rate schedules, dropout, L2, gradient clipping, and sequence models are outside these logged comparisons.

Conclusion: Performance is sensitive to tuning choices. More layers, more neurons, or a smaller learning rate do not consistently improve validation results.

Team contributions

Team member	Team Contributions
Ian Allish	**2–5:** Objective and approach, data, training, baseline
Joseph Falcone	**6–8:** Batch size, layer count, neuron count
Will Schneider	**9–11:** Activations, optimizers, learning rates
Noah Hoogerheide	**12–14:** Findings, limitations, team roles

AI assistance: Codex helped revise this presentation.